LIBBY — MANUSCRIPTS INVENTORY

Manuscripts considered — pancreatic-recurrent-kras-g12r-m8f3

Master inventory: every paper reviewed in this case

Generated 2026-05-12

Manuscripts considered — pancreatic-recurrent-kras-g12r-m8f3

Master inventory: 16 clinical (12 included, 4 considered & excluded) + 22 pre-clinical (19 included, 3 considered & excluded) + 22 additional trial publications/registrations. One row per manuscript, sorted by year (newest first). Excluded rows are papers a Libby agent reviewed and chose NOT to feed to the board, with the exclusion reason captured.

Report		Type	Inclusion	Indication / model	Design	n	Effect size	Variance	Toxicities (type, n/N, rate)	Case fit	Notes
(2026) — New England Journal of Medicine	PMID 42090791	clinical	included	Previously treated advanced RAS-mutated pancreatic ductal (PDAC) (2L+); RMC-6236-001 (NCT05379985) phase 1/2 PDAC cohort; 168 PDAC patients dosed at 300 mg or lower; second-line RAS G12 subgroup analyzed at the 300-mg dose (n=26). G12X variants enrolled include G12D, G12V, G12R, and Q61H — the patient's G12R is in scope.	Open-label multicenter phase 1/2 and	168	35 % — ORR (RECIST 1.1), median PFS, median OS, median DoR in 2L RAS G12 PDAC at 300 mg	—	any treatment-related AE (n=168, 96%); any treatment-related AE G>=3 (n=168, 30%); rash (n=127, 91%); rash G>=3 (n=127, 8%); diarrhea (n=127, 48%); diarrhea G>=3 (n=127, 2%); nausea (n=127, 43%); stomatitis (n=127, 31%); stomatitis G>=3 (n=127, 3%); vomiting (n=127, 31%); fatigue (n=127, 20%)	strong	Pivotal NEJM publication of the RMC-6236-001 PDAC cohort. The ORR 35% / mPFS 8.5 / mOS 13.1 in 2L RAS G12 at 300 mg is the load-bearing efficacy anchor for this case — G12R sits inside the G12 pooled cohort. G12R-specific subgroup ORR was not separately published; the field reads G12X as a single estimand pending phase 3 readout. Allele-resolved breakdown is the open question.

BMS/BMS (2026) — —	—	trial	included	/ PRMT5 inhibitor) ± or	Solid tumors (GBM, melanoma, NSCLC, PDAC) with homozygous MTAP deletion (2L+); Homozygous MTAP deletion	Phase 2 platform	—	— — ORR, PFS	—	—	partial	(registration only — no peer-reviewed publication yet; toxicity table not extracted)
(2025) — Nature Medicine	PMID 40790272	clinical	included	ELI-002 2P (amphiphile mKRAS peptide vaccine) — final report	Resected PDAC and CRC with MRD positivity post locoregional therapy (AMPLIFY-201 final follow-up) (adj); Final report at 19.7 months median follow-up; n=25 (20 PDAC, 5 CRC); KRAS G12D / G12R targeted.	Phase 1 open-label single-arm	25	— — Final RFS, OS, sustained T-cell responses	—	Maintained tolerability at extended follow-up; 71% of evaluable patients induced both CD4 and CD8 KRAS-specific T-cell responses.	partial	Final 2025 readout. Strengthens the vaccine class signal in MRD-positive resected disease and supports the ongoing 7-peptide successor (ELI-002 7P, NCT05726864). Overt recurrence sits outside the trial's enrollment window for this case. — (primary endpoint not extracted)

(2025) — Journal of Clinical Oncology	PMID 40644648	clinical	included	(RAF/MEK clamp) + defactinib (FAK inhibitor)	Recurrent low-grade serous ovarian cancer (LGSOC), KRAS-mutant and wild-type (RAMP 201) (2L+); Recurrent LGSOC after ≥1 prior systemic therapy; KRAS-mutant ORR pulled separately. Cross-tumor: not PDAC but the load-bearing efficacy precedent for the RAF/MEK + FAK clamp combo in a KRAS-mutant disease.	Phase 2 multicenter open-label	—	44 % ORR in KRAS-mutant LGSOC — Confirmed ORR by BICR; mPFS; mDoR; subset analyses by KRAS status	—	Manageable AE profile; most patients continued therapy. Class effects include rash, edema, CK elevation, and reversible ocular events.	FDA accelerated approval for KRAS-mutant recurrent LGSOC based on this readout. Cross-tumor precedent for the backbone that RAMP-205 (NCT05669482) is testing in 1L PDAC.

(2025) — Clinical Cancer Research	PMID 39437014	clinical	included	palbociclib	advanced solid tumors with CDK4 or CDK6 amplification (NCI-MATCH subprotocol Z1C) (2L+); 38 eligible/treated patients with confirmed CDK4 or CDK6 amplification (≥7 copies). 25 patients in the primary analysis cohort. Pancreatic cancer was not the dominant histology; the read-out is for the CDK4/6 amplification call, not CDKN2A loss.	Phase 2 basket trial arm (single-arm)	38	4 % ORR — ORR (primary), mPFS, mOS	—	neutropenia G>=3 (n=38)	weak	Negative basket readout for palbociclib in CDK4/6-amplified tumors (ORR 4%). This is the closest published precedent for off-label palbociclib in a CDKN2A-loss / CCND3 case — the message it sends is that monotherapy CDK4/6 inhibition does not deliver in unselected biomarker-positive solid tumors. CDKN2A-loss enrichment (the patient's actual feature) was not the gating biomarker here; CAPTUR Group 8 and NCI-MATCH Z1F are the more relevant arms but those readouts are not yet published.

(2025) — Cancer Research	doi:10.1158/0008-5472.CAN-25-0428	included	KRAS PDAC distinctive features (ERK / MAPK and tumor	Retrospective molecular and transcriptomic analysis of 2,447 PDAC tumors with paired organoid and mouse-model validation (KrasG12R vs KrasG12D ductal organoids; orthotopic implants)	G12R produces a weaker ERK / MAPK output than other G12 alleles, with reduced downstream activation. The functional is increased relative on residual MAPK signaling and a distinctive that may behave differently under direct RAS-GTP blockade.	with WT, per	moderate — KrasG12R PDAC organoids showed decreased migration and improved survival in orthotopic models versus G12D, alongside a distinctive low-ERK / transcriptional signature. The G12R subset's lower baseline MAPK flux predicts a wider therapeutic window for active-state RAS inhibition than the high-flux G12D class.	vs KRAS G12D and KRAS wild-type organoids; orthotopic injection comparators; translatability: med	n/a (preclinical)	strong	claim rests partly on bulk-RNA deconvolution, not single-cell validation across all samples. The lower-ERK phenotype is a relative not absolute observation.

Revolution Medicines (2025) — —	—	trial	included	zoldonrasib + +	Advanced / metastatic solid tumors with RAS mutation (NSCLC, CRC; PDAC eligibility per protocol) (2L+); RAS mutation; RAS(ON) inhibitor backbone with ivonescimab PD-1×VEGF bispecific	Open-label and expansion combining RAS(ON) inhibitors with	—	— — Safety, RP2D, ORR	—	—	partial	(registration only — no peer-reviewed publication yet; toxicity table not extracted)
(2025) — —	—	trial	included	ASP5834 pan-KRAS inhibitor	Locally advanced / metastatic solid tumors with KRAS mutations or amplification — NSCLC, PDAC, CRC (2L+); KRAS G12V, G12D, G12C, G12R, G12A, G13D, or KRAS amplification (copy number ≥4)	Phase 1 IV + combination with in CRC	364	— — Safety, RP2D, ORR	—	—	strong	(registration only — no peer-reviewed publication yet; toxicity table not extracted)
(2025) — —	—	trial	included	SBRT + + defactinib	Borderline resectable / locally advanced PDAC (neoadj); —	phase 2 (5:1) with SBRT plus FAK / RAF-MEK inhibition	36	— — Resectability, pathologic response, PFS	—	—	weak	(registration only — no peer-reviewed publication yet; toxicity table not extracted)

ADOPT (2025) — —	—	trial	included	Therapy from a menu including abemaciclib (CDK4/6), agents, PARPi where indicated	Advanced PDAC selected by patient-derived organoid (PDO) drug sensitivity (2L+); PDO selection; genomic profiling overlay	Adaptive platform	—	— — ORR, PFS, concordance	—	—	partial	(registration only — no peer-reviewed publication yet; toxicity table not extracted)
(2025) — —	—	trial	included	(KRAS +	Previously untreated KRAS G12D-mutated metastatic PDAC (DAWN-303) (1L); KRAS G12D	double-blind phase 3	—	— — OS, PFS	—	—	none	(registration only — no peer-reviewed publication yet; toxicity table not extracted)
(2025) — —	—	trial	included	ASP3082 KRAS degrader) + or NALIRIFOX	First-line KRAS G12D-mutated metastatic PDAC (1L); KRAS G12D	Phase 3 randomized	—	— — OS, PFS	—	—	none	(registration only — no peer-reviewed publication yet; toxicity table not extracted)

(2024) — Nature Medicine	PMID 38195752	clinical	included	ELI-002 2P (amphiphile mKRAS peptide vaccine)	Resected PDAC and CRC with ctDNA / tumor-marker MRD positivity after locoregional therapy (AMPLIFY-201) (adj); MRD-positive after curative-intent surgery and standard adjuvant therapy; 25 patients (20 PDAC, 5 CRC); KRAS G12D or G12R required. The patient's G12R is in scope, but the patient is overt-recurrent rather than MRD-only.	Phase 1 open-label single-arm	25	84 % with T-cell response — Safety, (T-cell response), response, relapse-free survival	—	dose-limiting toxicity G>=3 (0/25, 0%); injection-site reaction (n=25)	partial	G12R explicitly in the 2-peptide panel. Cohort is MRD-positive resected disease, not overt-recurrent — case fit is partial. Predecessor to ELI-002 7P (NCT05726864).

(2024) — Journal of Clinical Oncology (ASCO 2024 abstract 4140)	doi:10.1200/JCO.2024.42.16_suppl.4140	clinical	included	+ defactinib +	Treatment-naive metastatic PDAC, KRAS-mutant or wild-type (RAMP 205) (1L); Phase 1b/2; 41 patients dosed across 4 dose levels as of May 2024; KRAS activating mutation eligibility includes G12X / G13 / Q61. 1L untreated metastatic — patient is recurrent post-adjuvant FOLFIRI, so line fit depends on whether adjuvant FOLFIRI counts as prior systemic exposure in this protocol.	Phase 1b/2 open-label dose-finding plus expansion	41	83 % ORR at dose level 1 (5/6) — Confirmed ORR (early signal at dose level 1)	—	Manageable across four dose cohorts; details pending full publication. FDA orphan drug designation granted on the back of this signal.	partial	ASCO 2024 abstract; full publication pending. 5-of-6 confirmed PRs at dose level 1 is striking but small. RAMP 205 trial registry NCT05669482 corresponds to the trials.jsonl row; included here for the efficacy preview that would weight any discussion.

(2024) — Nature	PMID 38589574	included	Pan-cancer cell-line panel (>700 lines) with KRAS / NRAS / HRAS variants including KRAS G12R; subcutaneous xenografts across G12D, G12V, G12R, G12C, Q61 backgrounds	tri-complex with cyclophilin A that binds active RAS-GTP and sterically blocks across mutant and wild-type KRAS, NRAS, HRAS. Broad RAS(ON) blockade is the relevant mechanism for KRAS G12R, which is GTP-loaded and outside the reach of GDP-state covalent G12C drugs.	of (n not in	strong — RMC-6236 inhibited proliferation across G12X-mutant lines including G12R-bearing pancreatic and lung models, with on-target ERK suppression at sub-100 nM. Xenograft regression was achieved across multiple G12 alleles at tolerated oral exposures, and the molecule maintained activity in lines where adagrasib or sotorasib had no effect.	vs vehicle and comparators (e.g. sotorasib in G12C lines); translatability: high	n/a (preclinical)	strong	G12R-specific subgroup not separately quantified; activity is read from pooled G12X panels and a small number of G12R lines. Toxicology of pan-RAS blockade rests on a residual wild-type-RAS pharmacology window that is tighter in some tissues than others.

(2024) — Nature	PMID 38588697	included	tool compound RMC-7977 (RAS(ON)	PDAC translational suite: human and murine cell lines, patient-derived organoids, human PDAC explants, subcutaneous and orthotopic CDX / PDX, syngeneic allografts, and the autochthonous KPC Pdx1-Cre) GEMM	Active-state RAS(ON) tri-complex inhibitor selective for GTP-bound mutant and wild-type KRAS, NRAS, HRAS. Translates to PDAC because virtually all PDAC tumor cells depend on RAS-GTP signaling, and the differential tolerance reflects oncogene addiction rather than allele identity.	n per arm n per	strong — Treated PDAC tumors showed waves of apoptosis with sustained proliferative arrest, while normal tissues had only transient and apoptosis-free decreases in proliferation. In KPC mice RMC-7977 produced the longest survival extension recorded for that model and complete radiologic responses in a subset, with relapse linked to MYC copy-number gain.	vs vehicle; chemotherapy comparators in some arms; translatability: high	n/a (preclinical)	partial	KPC carries G12D rather than G12R, so the GEMM data are class-level rather than allele-specific. RMC-7977 is the tool compound; daraxonrasib (RMC-6236) is the clinical translate.

Jiang/Singh (2024) — Cancer Discovery	PMID 38593348	included	translational evaluation	Translational pan-RAS panel: cell lines, patient-derived organoids, and CDX / PDX models stratified by KRAS G12 allele (G12C, G12D, G12V, G12R) across PDAC, NSCLC, and CRC	RAS(ON) inhibition with tri-complex. The translational companion to activity stratified by G12 variant, including the rarer G12R allele relevant to this case.	n not in	strong — RMC-6236 produced ERK pathway suppression and tumor regression across G12C, G12D, G12V, and G12R-mutant PDAC models, with G12R-bearing lines responding within the same exposure window as other G12 alleles. engagement (DUSP6, SPRY suppression) and tumor regression were concordant across alleles.	vs vehicle; KRAS inhibitors for translatability: high	n/a (preclinical)	strong	G12R cohort within the translational panel is smaller than G12D / G12V cohorts; allele-resolved durability data are read from a limited number of lines.

(2024) — JCI Insight	PMID 39288262	included	NIS793 / antibody plus chemo + anti-PD-1 (preclinical rationale)	Orthotopic and autochthonous murine PDAC models (KPC, KPP) with TGF-beta blockade by 1D11 (murine surrogate for NIS793)	TGF-beta drives fibroblast activation, ECM deposition, and T-regulatory cell recruitment that together build the barrier in PDAC. Antibody blockade of TGF-beta-1 and TGF-beta-2 reverses the stromal program and re-permits drug penetration and T-cell entry.	per arm	moderate — TGF-beta blockade reduced fibroblast activation and ECM deposition in metastatic PDAC models, and added incremental tumor control on top of chemotherapy and anti-PD-1. The mechanism anchored the daNIS-2 / daNIS-3 NIS793 phase 3 design.	vs Vehicle; anti-PD-1 alone; translatability: low	n/a (preclinical)	weak	Novartis discontinued NIS793 PDAC development in 2023 after the daNIS-2 phase 3 readout did not meet endpoints; the preclinical signal did not survive clinical translation. Included as a foundational mechanism reference for the CAF / stroma-targeting axis the board may revisit with next-generation TGF-beta-trap or LRRC15-targeting agents.

Revolution Medicines (2024) — —	—	trial	included		Previously treated metastatic pancreatic ductal (2L+); RAS mutation (KRAS / NRAS / HRAS codon 12/13/61); G12R within scope	open-label, phase 3 versus choice SoC	—	— — OS, PFS, ORR	—	—	partial	(registration only — no peer-reviewed publication yet; toxicity table not extracted)
Revolution Medicines (2024) — —	—	trial	included	+ chemo or RMC-9805 in	GI solid tumors (PDAC, CRC) with RAS mutation (1L and 2L+); RAS mutation; subprotocols for G12D / G12X / G12C variants	Open-label multi-arm platform of RAS(ON) inhibitors as and with	—	— — Safety, RP2D, ORR, PFS	—	—	strong	(registration only — no peer-reviewed publication yet; toxicity table not extracted)
Pfizer/Pfizer (2024) — —	—	trial	included	pan-KRAS inhibitor; with gem/nab, FOLFOX, cetuximab,	Advanced solid tumors with KRAS mutations (G12C, G12D, G12V, G12R, G12S, G13D, Q61H) — NSCLC, CRC, PDAC (2L+ (mono); 1L (with gem/nab combo arm)); KRAS pan-G12 / G13D / Q61H, including G12R	Phase 1 open-label + and	330	— — Safety, RP2D, ORR, PFS	—	—	strong	(registration only — no peer-reviewed publication yet; toxicity table not extracted)

(2024) — —	—	trial	included	ELI-002 7P amphiphile KRAS peptide vaccine +	Resected PDAC and other KRAS / NRAS mutant solid tumors in the adjuvant minimal residual disease setting (adj); KRAS / NRAS G12D, G12R, G12V, G12A, G12C, G12S, G13D — 7-peptide panel	Phase 1/2 with adjuvant design and randomized phase 2 expansion (2:1)	135	— — survival, ctDNA reduction,	—	—		partial	(registration only — no peer-reviewed publication yet; toxicity table not extracted)
Academic consortium (2024) — —	—	trial	included	MEK inhibitor + (NTO-RAS)	RAS-mutant solid tumors including PDAC (2L+); RAS mutation	Phase 2 multi-cohort	—	— — ORR, PFS	—	—		partial	(registration only — no peer-reviewed publication yet; toxicity table not extracted)
(2024) — —	—	trial	included	(AMG 193 / PRMT5 inhibitor) + gem/nab or or RMC-6236	Advanced GI / biliary / pancreatic cancer with homozygous MTAP deletion (1L and 2L+); Homozygous MTAP deletion (commonly co-deleted with CDKN2A)	Phase 1b combination	—	— — Safety, RP2D, ORR	—	—		partial	(registration only — no peer-reviewed publication yet; toxicity table not extracted)

(2023) — New England Journal of Medicine	PMID 36546651	clinical	included	sotorasib (AMG 510)	Pretreated advanced KRAS pancreatic ductal (CodeBreaK 100 PDAC cohort) (2L+); 38 patients with KRAS G12C-mutated PDAC; median 2 prior lines (range 1-8). G12C is a different allele from the patient's G12R — sotorasib does not bind G12R.	Phase 1/2 single-arm pooled (phase 1 n=12 + phase 2 n=26)	38	21 % — ORR (centrally reviewed)	95% CI 10–37	any treatment-related AE (16/38, 42%); any treatment-related AE G>=3 (6/38, 16%); diarrhea (n=38); fatigue (n=38)	Mechanism-class context only — sotorasib is G12C-selective and does not act on KRAS G12R. The 21% ORR establishes proof-of-concept that allele-selective covalent KRAS blockade works in PDAC, which is what anchors the pan-RAS / pan-G12X rationale for a G12R patient. Tag: case_match cross_tumor_only flags the mutation-class mismatch even though the indication matches.

(2023) — Journal of Clinical Oncology	PMID 37099736	clinical	included	adagrasib (MRTX849)	Advanced solid tumors with KRAS G12C mutation, excluding NSCLC and CRC (KRYSTAL-1 cohort) (2L+); 63 treated patients; pancreatic n=21, biliary tract n=12, other GI / gyne. G12C-only — not relevant to the patient's G12R at the molecular level.	Phase 1/2 single-arm cohort	63	35.1 % — ORR per investigator	—	any treatment-related AE (n=63, 96.8%); any treatment-related AE G>=3 (n=63, 27%); nausea (n=63, 49.2%); diarrhea (n=63, 47.6%); fatigue (n=63, 41.3%); vomiting (n=63, 39.7%)		PDAC subset ORR 33.3% (7/21) and biliary 41.7% (5/12) — consistent with sotorasib in the same indication. G12C-selective; off-axis for a G12R patient. Logged for the allele-selective-KRAS class precedent that the pan-RAS programs are extending to non-G12C alleles.
(2023) — Lancet	PMID 37708904	clinical	excluded — 1L that does not target the patient's molecular features. per Libby's rule.	NALIRIFOX (NAPOLI 3)	Treatment-naive metastatic pancreatic first-line	open-label phase 3 vs gem/nab	—	—	—	—	none	Excluded: Standard-of-care 1L chemotherapy that does not target the patient's molecular features. Out-of-scope per Libby's feature-targeting rule. — OS 11.1 vs 9.2 mo (HR 0.83, p=0.036) — new 1L reference regimen for mPDAC. The patient has had FOLFIRI adjuvant (irinotecan-exposed) but is oxaliplatin-naive, which keeps this regimen on the SoC table at the treating team's discretion.

(2023) — Cancer Discovery	PMID 37552839	included	MRTX1719 PRMT5 inhibitor)	HCT116 isogenic MTAP-WT vs MTAP-deleted; broad cell-line panel of MTAP-deleted vs MTAP-WT tumors; xenograft models including MTAP-deleted PDAC, NSCLC, melanoma	Selective binding to the complex, leveraging the elevated MTA pool in cells. The cooperative mechanism produces >70-fold selectivity for over cells, sparing normal tissue and avoiding the hematologic toxicity of PRMT5 inhibitors.	cell in per arm	strong — MRTX1719 inhibited SDMA methylation and cell viability in MTAP-deleted lines with >70-fold selectivity, and produced tumor regression in MTAP-deleted xenografts at well-tolerated doses. Hematologic and bone-marrow effects, the limiting feature of PRMT5 inhibitors, were not observed at efficacious doses.	vs MTAP-WT matched lines; vehicle in xenografts; translatability: high	n/a (preclinical)	partial	PDAC-specific xenografts represented a minority of the in-vivo panel; the bulk of in-vivo data is in NSCLC, mesothelioma, and melanoma models. Clinical translate for this case requires confirmed MTAP co-deletion.

(2023) — Cancer Research	PMID 36346366	included	CDK4/6 inhibition plus MEK / ERK inhibition (palbociclib + trametinib or ulixertinib)	KRAS-mutant PDAC cell lines (MIA PaCa-2, Panc-1, AsPC-1, Capan-1) and matched patient-derived organoids; subcutaneous xenografts of KRAS-mutant PDAC	ERK signaling and CDKN2A loss converge on RB inactivation. Concurrent CDK4/6 and ERK / MEK blockade prevents the of cyclin D, MYC, and signaling that limits single-agent palbociclib in PDAC.	Six per arm	strong — Combined CDK4/6 and ERK / MEK inhibition produced synergistic suppression of proliferation in all six PDAC organoid models tested and tumor regression in xenografts where either agent alone produced cytostasis. The combination anchored the perioperative palbociclib + binimetinib trial and the ulixertinib + palbociclib study	vs vehicle; single-agent palbociclib; single-agent trametinib or ulixertinib; translatability: med	n/a (preclinical)	strong	Most lines are KRAS G12D / G12V; G12R-specific organoid data are absent. The synergy logic still applies because both axes converge on RB through the same effector chain, but G12R's lower baseline ERK flux is an open variable.

(2023) — —	—	trial	included	(VS-6766, RAF/MEK clamp) + defactinib (FAKi) + gemcitabine +	Previously untreated metastatic PDAC with KRAS activating mutation (1L); KRAS activating mutation (any G12X including G12R; G13D; Q61)	Phase 1b/2a open-label	40	— — Safety, RP2D, ORR, PFS	—	—	partial	(registration only — no peer-reviewed publication yet; toxicity table not extracted)
Mirati/Mirati (2023) — —	—	trial	included	MRTX1133 (KRAS	Advanced solid tumors with KRAS G12D mutation (2L+); KRAS G12D	Phase 1/2 (phase 2 not initiated)	63	Terminated for formulation issues — Safety, RP2D, ORR	—	—	none	(registration only — no peer-reviewed publication yet; toxicity table not extracted)
(2022) — Nature	PMID 36028483	clinical	included	durvalumab + +	Treatment-naive metastatic PDAC (CCTG PA.7) (1L); 180 randomized to chemo + dual ICI vs chemo alone; unselected for biomarker. The patient is recurrent rather than 1L untreated, but the negative readout is for any ICI-combination discussion.	open-label phase 2	180	— — Overall survival (primary)	p=0.72	lymphocyte elevation (n=180)	weak	Negative RCT for dual ICI added to chemo in unselected metastatic PDAC. The subgroup signal in KRAS-wildtype patients is hypothesis-generating only. For an MSS / sub-10-TMB G12R patient, this is the load-bearing negative evidence for adding biomarker-agnostic ICI to chemo. — (primary endpoint not extracted)

(2022) — Cells	PMID 35269414	included	APR-246 / (mutant TP53 reactivator)	PDAC cell lines PANC-28 (TP53-null) and MIA PaCa-2 (TP53 R248W isogenic wild-type TP53 reintroduction	APR-246 covalently modifies cysteine residues in mutant p53 to restore wild-type and activity. The mechanism is selective for missense (especially unstable hotspot) mutants; pure or truncating TP53 variants lack a target to reactivate.	n=3 per	moderate — APR-246 sensitized MIA PaCa-2 (R248W PDAC cells to combination cytotoxics and derivatives, with effect size dependent on the presence of mutant or reintroduced wild-type TP53 protein. TP53-null PANC-28 cells were not sensitized, confirming target-class specificity.	vs Vehicle; matched parental lines without WT-TP53 reintroduction; translatability: low	n/a (preclinical)	weak	In-vitro only; no PDAC xenograft data. Activity requires a missense / conformationally unstable mutant TP53 protein. The case's TP53 variant class is annotated as inactivating without missense specification; APR-246 is off-target if the variant is a truncation or splice-site allele.

(2022) — Cancer Discovery	PMID 36216931	excluded — by design; does not bind G12R. Reviewed because the preclinical PDAC efficacy is striking (deep regressions in implantable and KPC) but the allele specificity means the mechanism is off-target for this patient. Logged here so the audit trail captures that the class precedent was examined and rejected on allele grounds; the pan-RAS RMC-6236 / RMC-7977 rows are

the on-axis

evidence for

G12R.	MRTX1133 (KRAS inhibitor)	—	—	—	—	—	n/a (preclinical)	none	Excluded: G12D-selective by design; does not bind G12R. Reviewed because the preclinical PDAC efficacy is striking (deep regressions in implantable and autochthonous KPC) but the allele specificity means the mechanism is off-target for this patient. Logged here so the audit trail captures that the class precedent was examined and rejected on allele grounds; the pan-RAS RMC-6236 / RMC-7977 rows are the on-axis evidence for G12R.

Revolution Medicines (2022) — —	—	trial	included	oral	Advanced solid tumors with mutations at codons 12, 13, or 61 — includes PDAC, NSCLC, CRC cohorts (2L+); KRAS G12X (including G12R), NRAS, HRAS at codons 12/13/61; PDAC RAS-WT also eligible	Multicenter open-label + (FIH)	—	— — Safety, RP2D, ORR (RECIST 1.1), PFS	—	—	strong	(registration only — no peer-reviewed publication yet; toxicity table not extracted)
(2022) — —	—	trial	included	trametinib 2 mg/day + 1200 mg/day (PaTcH)	Metastatic refractory pancreatic cancer (2L+); —	Single-arm phase 2	20	Terminated for futility — ORR, PFS	—	—	weak	(registration only — no peer-reviewed publication yet; toxicity table not extracted)
(2022) — —	—	trial	included	MEK (MEKi + HCQ, MEKi + EGFRi, etc.)	Advanced KRAS G12R altered PDAC (2L+); KRAS G12R	registry of MEK-based combination therapy	—	— — ORR, PFS, OS	—	—	strong	(registration only — no peer-reviewed publication yet; toxicity table not extracted)

(2021) — Journal of Clinical Oncology	PMID 33970687	clinical	included	rucaparib (Rubraca)	advanced PDAC with germline or somatic BRCA1/2 or PALB2 pathogenic variant (maintenance); 46 enrolled, 42 evaluable; germline BRCA1 n=7, BRCA2 n=27, PALB2 n=6; somatic BRCA2 n=2. Broader than POLO by including PALB2 and somatic variants.	Phase 2 single-arm	46	59.5 % PFS rate at 6 months — PFS at 6 months (primary), ORR, OS	—	anemia (n=46, 74%); anemia G>=3 (n=46, 22%); nausea (n=46, 48%); nausea G>=3 (n=46, 2%); increased ALT (n=46, 47%); increased ALT G>=3 (n=46, 4%); fatigue (n=46, 45%); fatigue G>=3 (n=46, 4%); thrombocytopenia (n=46, 39%); thrombocytopenia G>=3 (n=46, 4%); dysgeusia (n=46, 37%)	partial	Extends the PARP signal beyond gBRCA to include sBRCA and PALB2; relevant for the case only if germline / somatic HRD testing returns positive. Currently not on file; gated by the germline panel in target_validation.jsonl. Safety data extracted from in-text Safety paragraph and Appendix Table A4 of JCO 2021;39:2497-2505.

(2021) — Journal for of Cancer	PMID 34376552	included	Amphiphile mKRAS peptide vaccine (ELI-002 preclinical lead)	C57BL/6 mice bearing syngeneic CT26 and MC38 colorectal allografts (KrasG12D) and Panc02-derived models; amphiphile peptide constructs with CpG-7909 adjuvant	lipid conjugation drives peptide trafficking to draining lymph nodes, where high-density antigen and CpG signaling expand CD8 and CD4 T cells. The strategy targets the KRAS peptide neoantigens (G12D and G12R variants) shared across PDAC and CRC.	n per arm	strong — Amphiphile delivery generated T-cell responses than soluble peptide and produced durable tumor control with cures in a fraction of KRAS-mutant syngeneic models. Memory T-cell expansion persisted for >100 days and conferred resistance to tumor rechallenge.	vs Soluble peptide vaccine; CpG alone; vehicle; translatability: med	n/a (preclinical)	partial	Murine syngeneic models are MSI-high relative to human PDAC and likely overstate immunogenic vaccine windows. The clinical ELI-002 AMPLIFY-201 readout in MRD-positive PDAC is the more conservative translatability anchor.

NCI/NCI (2021) — —	—	trial	included	olaparib	Resected PDAC with germline or somatic BRCA1/2 or PALB2 mutation (adj); Germline / somatic BRCA1, BRCA2, or PALB2 pathogenic variant	double-blind phase 2 (APOLLO)	152	— — Relapse-free survival (primary)	—	—	none	(registration only — no peer-reviewed publication yet; toxicity table not extracted)
(2020) — Journal of Clinical Oncology	PMID 31682550	clinical	included	(Keytruda)	MSI-H / dMMR previously treated noncolorectal advanced cancer (2L+); 233 enrolled across 27 tumor types; pancreatic n=22; MSI-H / dMMR required. The patient is MSS — from this indication. Logged to harden the MSS-PDAC ICI foreclosure call.	Multi-cohort open-label phase 2	233	18.2 % ORR in MSI-H pancreatic subgroup (4/22) — ORR overall and per-tumor (pancreatic subset)	—	any treatment-related AE G>=3 (34/233, 14.6%); discontinuation for immune-mediated AE (n=233, 5.2%)	none	The PDAC ORR (18%) is real only for MSI-H disease, which the patient does not have. PDAC subgroup mPFS 2.0 / mOS 4.0 in the responder-enriched MSI-H cohort gives a ceiling for what monotherapy ICI delivers even when biomarker-positive. The patient is MSS / TMB 4.1 — the door is closed.

Hobbs/Der (2020) — Cancer Discovery	PMID 31649109	included	KRAS G12R (informs pan-RAS targeting rationale)	Isogenic RIE-1 rat intestinal epithelial cells and PDAC lines (HPAC, Capan-1) with knock-in of KRAS G12D vs G12V vs G12R; biochemical and structural characterization	Structural analysis showing the G12R arginine substitution partially unfolds helix alpha-2 of switch II, displacing Q61 and forming a direct interaction with E62 / T35. The resulting altered switch-II surface impairs PI3K p110-alpha binding and abolishes induction relative to G12D / G12V, while preserving Raf	moderate — KRAS G12R is functionally distinct from other G12 alleles: stable expression of G12D or G12V increased whereas G12R did not, and G12R-PDAC lines showed selective dependence on alternative scavenging pathways. The work anchors why allele-class context matters and why G12R sits in the RAS(ON) tri-complex window rather than the GDP-state covalent G12C window.	vs KRAS wild-type and other G12 variants (G12D, G12V) as comparators; translatability: med	n/a (preclinical)	strong	Biology paper, not a therapeutic efficacy paper. Provides the mechanistic basis for selecting pan-RAS over covalent G12C-class drugs in this case; does not by itself predict on-treatment response magnitude.

(2020) — Journal of Medicinal Chemistry	PMID 31790586		included	olaparib plus disruptor (preclinical extension)	Capan-1 and BxPC-3 PDAC cell lines; isogenic comparisons	PARP inhibition collapses repair, generating lesions that require homologous for resolution. BRCA1/2- or PDAC cells lack this repair channel, producing the vulnerability that drove the olaparib FDA approval in PDAC.		strong — Capan-1 (BRCA2-null) PDAC cells were >100-fold more sensitive to olaparib than BxPC-3 cells, recapitulating the canonical phenotype. Combining olaparib with a disruptor extended the sensitivity into lines.	vs Vehicle; single-agent olaparib; single-agent disruptor; translatability: high	n/a (preclinical)	weak	Mechanism is irrelevant to this case unless the germline panel returns a BRCA1, BRCA2, or PALB2 hit. Included as the mechanism-of-action reference behind the POLO and APOLLO trial design, which the board may pivot to if germline testing turns positive.
NCI/NCI (2020) — —	—	trial	included	palbociclib	Rare solid tumors with actionable genomic alteration — platform (2L+); CDKN2A mutation (palbociclib arm)	platform for rare tumors	—	— — ORR, clinical benefit	—	—	weak	(registration only — no peer-reviewed publication yet; toxicity table not extracted)

(2019) — New England Journal of Medicine	PMID 31157963	clinical	included	olaparib (Lynparza)	Germline metastatic PDAC, maintenance after first-line platinum-based chemotherapy (POLO) (maintenance); 154 randomized; gBRCA1/2 pathogenic variant; no progression after ≥16 weeks of first-line platinum. The patient has no germline data yet and was treated with adjuvant FOLFIRI (no platinum exposure), so both biomarker and gates are open.	double-blind phase 3	154	0.53 HR — PFS (primary)	95% CI 0.35–0.82; p=0.004	any treatment-related AE G>=3 (n=92, 40%); any treatment-related AE G>=3 (n=62, 23%); discontinuation for AE (n=92, 5%)	partial	Pivotal RCT driving FDA approval of olaparib maintenance in gBRCA-mutant metastatic PDAC. The case requires both germline testing (currently outstanding) and platinum induction (currently not on file) to unlock this pathway.

Bryant/Der (2019) — Nature Medicine	PMID 30833752	included	ERK / MEK inhibition plus autophagy blockade (trametinib + chloroquine /	Panel of KRAS-mutant PDAC cell lines (MIA PaCa-2, Pa01C, Pa02C, Pa14C, Pa16C); patient-derived xenografts; KPC GEMM-derived organoids and allografts	KRAS or ERK suppression drives autophagy that protects PDAC cells from metabolic stress. Combined ERK / MEK and autophagy blockade collapses both cytosolic and bioenergetic adaptation, producing synthetic lethal vulnerability that single-agent MEK or autophagy targeting does not deliver.	3-6 n per arm	strong — Combined ERKi or trametinib with chloroquine produced near-complete tumor regression across PDAC cell line xenografts and models, where single agents at best slowed growth. The combination also extended survival in KPC mice and synergized in syngeneic CT26 KRAS-mutant CRC allografts.	vs vehicle; single-agent SCH772984 (ERKi) or chloroquine alone; translatability: med	n/a (preclinical)	partial	Clinical translation has been mixed: PaTcH (trametinib + HCQ) terminated for futility despite the preclinical signal. The dose-intensity gap between mouse HCQ exposure and tolerable human dosing is the working hypothesis for the disconnect.

(2019) — Nature Medicine	PMID 30833748	included	MEK inhibition plus autophagy blockade (trametinib + HCQ; companion to Bryant 2019)	KRAS-mutant PDAC cell lines; patient-derived xenografts; KPC GEMM; melanoma and CRC RAS / RAF-mutant lines for cross-tumor validation; one human case report of a metastatic PDAC patient treated with trametinib + HCQ	inhibition induces a protective autophagy program that buffers metabolic stress in RAS-driven cancers. lysosomal disruption blocks this adaptive response and converts cytostasis into cytotoxicity.	of per n=1	strong — Combined MEK inhibition and autophagy blockade caused tumor regression in PDAC PDX and KPC models where either monotherapy delivered only growth slowing. A companion human case showed a metabolic response on FDG-PET and clinical benefit in a KRAS-mutant PDAC patient on trametinib plus HCQ.	vs vehicle; single-agent trametinib; single-agent HCQ; translatability: med	n/a (preclinical)	partial	Companion paper to Bryant 2019 (back-to-back NEJM-style co-publication). Same caveat applies: PaTcH futility tempers the clinical signal; the autophagy axis still drives the NTO-RAS basket and several next-generation ULK1 / VPS34 trials.

(2019) — Nature	PMID 31666701	excluded — covalent inhibitor; binds the cryptic GDP-state switch-II pocket via a reactive cysteine that G12R does not present. Off-target for this case. Reviewed because the preclinical foundation (anti-tumor immunity, synergy) is the class precedent the pan-RAS programs build on; the published efficacy is in NSCLC and CRC G12C models, not G12R PDAC. Logged for audit trail.	AMG 510 / sotorasib (KRAS covalent inhibitor)	—	—	—	—	—	n/a (preclinical)	—	Excluded: G12C-selective covalent inhibitor; binds the cryptic GDP-state switch-II pocket via a reactive cysteine that G12R does not present. Off-target for this case. Reviewed because the preclinical foundation (anti-tumor immunity, immune-checkpoint synergy) is the class precedent the pan-RAS programs build on; the published efficacy is in NSCLC and CRC G12C models, not G12R PDAC. Logged for audit trail.

(2018) — —	—	trial	included	palbociclib (CDK4/6 inhibitor)	Advanced solid tumors with actionable genomic alteration — basket (2L+); CDKN2A loss, CDK4 / CCND1 / SMARCA4 alteration (palbociclib arm — now closed)	Canadian Profiling and Targeted Agent Utilization Trial (CAPTUR) — phase 2 basket	—	— — ORR, clinical benefit	—	—	weak	(registration only — no peer-reviewed publication yet; toxicity table not extracted)
(2017) — —	—	trial	included	(APR-246) ±	esophageal / junction cancer with TP53 mutation (2L+); TP53 missense mutation / GOF reactivation target)	Phase 1b/2 single-arm (APROC)	5	— — Safety, ORR	—	—	none	(registration only — no peer-reviewed publication yet; toxicity table not extracted)

(2016) — Lancet	PMID 26615328	clinical	excluded — 2L that does not target the patient's molecular features. per Libby's rule. The patient already received FOLFIRI adjuvant, so prior irinotecan further limits the second-line nal-IRI option.	irinotecan + 5-FU/LV (NAPOLI-1)	Metastatic pancreatic second-line after therapy	phase 3 three-arm	—	—	—	—	none	Excluded: Standard-of-care 2L chemotherapy that does not target the patient's molecular features. Out-of-scope per Libby's feature-targeting rule. The patient already received FOLFIRI adjuvant, so prior irinotecan further limits the second-line nal-IRI option. — OS 6.1 vs 4.2 mo with nal-IRI+5FU vs 5FU, HR 0.67 — the comparator that the recurrent-PDAC SoC discussion would otherwise default to. Same scoping rationale as the other SoC chemo rows.

(2016) — Nature Medicine	PMID 27376576	included	FAK inhibition (VS-4718, defactinib precursor) plus checkpoint	Autochthonous KPC mouse model of PDAC Pdx1-Cre); human PDAC tissue correlative analysis	FAK in PDAC tumor cells drives a fibrotic, stromal program that excludes T cells. FAK inhibition collapses the stromal barrier, reduces MDSC and infiltration, and primes the tumor for response.	per arm in arm	strong — VS-4718 monotherapy doubled survival in KPC mice and reduced tumor fibrosis and MDSC infiltration; adding anti-PD-1 or anti-CTLA-4 produced sustained tumor regression in a model that is otherwise refractory to checkpoint blockade. The combination established the FAK + ICI rationale that drove the defactinib + and chemo trials in PDAC.	vs vehicle; anti-PD-1 or anti-CTLA-4 monotherapy; translatability: med	n/a (preclinical)	partial	KPC is G12D, not G12R; biology may differ in the lower-ERK G12R context (Singhi 2025). The downstream clinical FAK + ICI combinations have shown modest signals at best in unselected PDAC.

(2016) — Science	PMID 26912361	included	PRMT5 in MTAP / CDKN2A co-deleted cancers	Pan-cancer cell-line panel (>390 lines) including MTAP-deleted PDAC, NSCLC, and glioma lines; isogenic MTAP knockout and reconstitution clones	MTAP loss (co-deleted with CDKN2A on 9p21 in ~80-90% of CDKN2A deletions) drives intracellular of (MTA), which partially inhibits PRMT5. cells thus operate near a PRMT5 functional threshold, creating a vulnerability to further PRMT5 blockade.	strong — MTAP deletion sensitized cell lines to PRMT5 knockdown across tumor histologies, and reconstitution of MTAP rescued the dependence. The work established the molecular basis for the PRMT5 inhibitor programs (MRTX1719, AMG 193, BMS-986504) now in clinical development for MTAP-deleted PDAC and other indications.	vs MTAP wild-type matched lines; MTAP-deleted lines; translatability: high	n/a (preclinical)	partial	Translates to this case only if MTAP is co-deleted with CDKN2A on the same allele; an MTAP IHC or NGS copy-number call is the gating measurement. First-generation PRMT5 inhibitors lacked selectivity; the clinical translate is the MTA-cooperative class (MRTX1719 etc.).

(2016) — Science	PMID 26912360	included	PRMT5 in cancers (Kryukov companion paper)	Project DRIVE / Achilles RNAi screens across hundreds of cancer cell lines; biochemical PRMT5 / MTA assays; MTAP-isogenic systems	that cancers accumulate MTA, which competes at the PRMT5 active site and reduces baseline PRMT5 methylation activity. The PRMT5 state is selectively sensitive to additional PRMT5 inhibition, the vulnerability that defines the inhibitor class.	cell in	strong — Genome-wide shRNA screening identified PRMT5 as the top differential dependency in MTAP-deleted lines across multiple tumor histologies. The discovery, published back-to-back with Mavrakis 2016, anchors the bilateral preclinical foundation for the MTAP / PRMT5 class.	vs matched lines; in-vitro PRMT5 enzymatic assays with and without MTA; translatability: high	n/a (preclinical)	partial	Genetic dependence does not by itself indicate a tractable therapeutic window; first-generation PRMT5 inhibitors were too toxic, and the clinical translate required the MTA-cooperative chemical class to spare normal MTAP-proficient tissue.

(2014) — Oncotarget	PMID 25156567	included	palbociclib plus	Panel of 15 patient-derived PDAC primary explants; PDX-derived organoid models; established cell lines with CDKN2A loss	CDKN2A loss CDK4/6, which would predict palbociclib sensitivity, but pancreatic biology drives parallel pathways ERK feedback, IGF1R) that limit single-agent activity. with agents collapses the bypass routes.	moderate — Palbociclib suppressed proliferation in 14 of 15 primary PDAC explants in vitro, with the single resistant explant explained by RB1 loss. Single-agent durability in vivo was limited; combinations with PI3K, MEK, or IGF1R inhibitors converted cytostasis into sustained tumor control in PDX models.	vs vehicle; single-agent PD-0332991; translatability: med	n/a (preclinical)	partial	Single-agent CDK4/6 inhibition has subsequently underperformed in CDKN2A-altered solid-tumor baskets (NCI-MATCH Z1C palbociclib ORR 4%; CDK4/6-amplified TAPUR PDAC arm). Translates as combination-strategy rationale rather than monotherapy support.

(2014) — Cancer Research	PMID 24986516	included	palbociclib plus IGF1R inhibition	p16INK4A PDAC cell lines (Panc1, MIA PaCa-2 derivatives, BxPC-3); subcutaneous xenografts	PDAC retains active IGF1R / mTORC1 signaling that drives IGF1R / IR co-inhibition collapses the mTORC1 escape route; substitution of an mTOR inhibitor broadens the sensitivity envelope.	per arm	strong — BMS-754807 plus PD-0332991 strongly suppressed proliferation in p16-deficient PDAC lines in-vitro and produced tumor regression in xenografts where either agent alone produced only cytostasis. Sensitivity correlated with mTORC1 suppression, and rapalog substitution recapitulated the synergy.	vs vehicle; single-agent PD-0332991; single-agent BMS-754807; translatability: med	n/a (preclinical)	partial	Cell-line dominated panel without GEMM validation; IGF1R-class drugs have not delivered in clinical PDAC trials. The mechanism remains the cleanest rationale for CDK4/6 + mTOR / PI3K combinations in CDKN2A-loss PDAC.

Von (2013) — New England Journal of Medicine	PMID 24131140	clinical	excluded — that does not target the patient's molecular features. per Libby's rule. Logged as the comparator context for the gem/nab combination arms in RAMP-205 and	+ gemcitabine (MPACT)	Metastatic pancreatic first-line	phase 3 vs gemcitabine	—	—	—	—	none	Excluded: Standard-of-care chemotherapy that does not target the patient's molecular features. Out-of-scope per Libby's feature-targeting rule. Logged as the comparator context for the gem/nab combination arms in RAMP-205 and RMC-GI-102. — OS 8.5 vs 6.7 mo, HR 0.72 — the second 1L PDAC standard. Same scoping rationale as the FOLFIRINOX row.

Feig/Fearon (2013) — PNAS	PMID 24277834	included	CXCL12 / CXCR4 blockade (AMD3100) plus anti-PD-L1 in PDAC	Autochthonous KPC Pdx1-Cre) PDAC GEMM; FAP-DTR transgenic depletion of FAP-positive fibroblasts	fibroblasts produce CXCL12 that coats PDAC tumor cells and excludes CD8 T cells, accounting for the canonical phenotype. CXCR4 antagonism rapidly admits T cells to the tumor bed, where anti-PD-L1 then drives tumor regression.	n per arm	strong — AMD3100 produced rapid T-cell accumulation in PDAC tumors and synergized with anti-PD-L1 to substantially reduce tumor burden, with effects lost when FAP-positive fibroblasts were depleted. The work established the axis as the dominant mechanism in KPC PDAC.	vs Vehicle; anti-CTLA-4 monotherapy; anti-PD-L1 monotherapy; translatability: med	n/a (preclinical)	partial	Clinical translation has been disappointing: motixafortide + cemiplimab and other CXCR4-axis combinations have produced modest signals in MSS PDAC. KPC is G12D, not G12R; the G12R phenotype may differ (Singhi 2025 suggests a baseline immune-enriched compartment).

(2011) — New England Journal of Medicine	PMID 21561347	clinical	excluded — for the indication whose mechanism does not target the patient's targetable features (KRAS G12R, CDKN2A loss, CCND3, MSS, TMB 4.1). per Libby's rule. Included as a one-line audit-trail entry so the reviewer can see it was reviewed and decided not to compile.	Metastatic pancreatic first-line (PRODIGE 4/ACCORD 11)	phase 3 vs gemcitabine	—	—	—	—	none	Excluded: Standard-of-care chemotherapy for the indication whose mechanism does not target the patient's targetable features (KRAS G12R, CDKN2A loss, CCND3, MSS, TMB 4.1). Out-of-scope per Libby's feature-targeting rule. Included as a one-line audit-trail entry so the reviewer can see it was reviewed and decided not to compile. — OS 11.1 vs 6.8 mo, HR 0.57 for FOLFIRINOX vs gemcitabine — landmark 1L PDAC benchmark. The treating team handles SoC chemo independent of Libby; logging only as a comparator-context anchor for the board's framing of what 'standard alternatives deliver'.

(2009) — Science	PMID 19460966	excluded — mechanism paper that motivated the entire FAP / CAF / field. Reviewed but not retained as a primary entry because the angle has not produced clinical translation (IPI-926 and vismodegib trials failed) and the more directly relevant preclinical anchors for this case are Jiang 2016 (FAK) and Feig 2013 (CXCL12). Logged to show the lineage was considered.	Olive 2009 Science / PDAC paper	—	—	—	—	—	n/a (preclinical)	—	Excluded: Foundational mechanism paper that motivated the entire FAP / CAF / stromal-targeting field. Reviewed but not retained as a primary entry because the hedgehog-pathway angle has not produced clinical translation (IPI-926 and vismodegib trials failed) and the more directly relevant preclinical anchors for this case are Jiang 2016 (FAK) and Feig 2013 (CXCL12). Logged to show the lineage was considered.